QSNN-NIDS: What to Change & Why

Consolidated handoff for your draft · results, code fixes, and paper-text suggestions from the independent re-run (UNSW-NB15 + CIC-IDS2017, seed 42)

Bottom line: Good news first — your core QSNN-NIDS numbers reproduce well on an independent, larger re-run (within 1–2 points on every metric), and the model now has full results on a second dataset (CIC-IDS2017), a complete noise-robustness study, and quantum baselines on both. That closes three of the six gaps your own draft's Limitations section flagged. This document tells you exactly what changed, what to fix in your script, and what to paste into the paper.

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1. Results: Your Draft vs. the Re-Run

Your draft draws 50,000 records from only the official UNSW-NB15 training partition and re-splits them (7,284-flow test set). The re-run instead combines the official training *and* test CSVs — your loader had a bug silently dropping the test file — and re-splits the full 200,000-row set (26,028-flow test set), same architecture and seed (42).

Metric	Your draft (n=7,284)	Re-run (n=26,028)	Delta
Accuracy	0.8585	0.8630	+0.0045
Precision (attack)	0.7904	0.8369	+0.0465
Recall (attack)	0.8981	0.8903	−0.0078
F1 (attack)	0.8408	0.8628	+0.0220
FPR	0.1698	0.1626	−0.0072
MCC	0.7190	0.7276	+0.0086
ROC-AUC	0.9397	0.9382	−0.0015
Mean spike rate	0.284	0.144	−0.140 (worth checking)
Latency/flow	3.55–3.79 ms	7.91 ms	hardware-dependent, not a bug

Takeaway for you: this is a reproducibility win — state it explicitly in the paper. Most QML-IDS papers never show a second run survives. Section 5 below flags the few deltas worth a sentence of explanation rather than silence.

2. Results: New Dataset — CIC-IDS2017

Full pipeline run end-to-end: 300,000 rows, all 8 release files (one had a latin1-encoding issue in your loader — fixed, see Section 4), QSNN + all 5 baselines + the 6-point noise sweep.

UNSW-NB15

Model	Family	Accuracy	Precision	Recall	F1	FPR	ROC-AUC
QSNN (ours)	Quantum	0.8630	0.8369	0.8903	0.8628	0.1626	0.9382
Random Forest	Classical	0.9048	0.8888	0.9180	0.9031	0.1076	0.9751
RBF-SVM	Classical	0.8966	0.8812	0.9088	0.8948	0.1148	0.9662
MLP (DNN)	Classical	0.8966	0.8826	0.9069	0.8946	0.1130	0.9715
QSVC (ZZ kernel)	Quantum	0.8640	0.8417	0.8707	0.8559	0.1418	0.9361
QCNN	Quantum	0.8749	0.8530	0.8957	0.8739	0.1446	0.9532

CIC-IDS2017

Model	Family	Accuracy	Precision	Recall	F1	FPR	ROC-AUC
QSNN (ours)	Quantum	0.9746	0.9187	0.9560	0.9370	0.0208	0.9939

Random Forest	Classical	0.9965	0.9877	0.9946	0.9912	0.0030	0.9997
RBF-SVM	Classical	0.9901	0.9603	0.9910	0.9754	0.0101	0.9983
MLP (DNN)	Classical	0.9910	0.9668	0.9882	0.9774	0.0084	0.9994
QSVC (ZZ kernel)	Quantum	0.9720	0.9400	0.9216	0.9307	0.0151	0.9717
QCNN	Quantum	0.9715	0.8967	0.9674	0.9307	0.0275	0.9953

Why this matters: the QSNN's ranking relative to every baseline is consistent across both datasets (classical ahead, QSNN closer to QCNN/QSVC) — that's a generalization claim your single-dataset draft couldn't make. It also posts its best absolute numbers (F1 0.937) on CIC-IDS2017, tracking the fact that every model finds CIC-IDS2017 an easier separation task.

QSNN Resource Footprint (your paper's actual differentiator)

Metric	UNSW-NB15	CIC-IDS2017
Qubits	8	8
Circuit depth / timestep	15	15
Total circuit depth (12 timesteps)	360	360
Two-qubit gates / timestep	32	32
Quantum parameters	96	96
Classical parameters	244	244
Inference latency	7.91 ms/sample (126 flows/s)	5.10 ms/sample (196 flows/s)
Spike rate / sparsity saving	14.4% / 85.6%	18.0% / 82.0%

Architecture is identical across datasets (same qubit count, depth, params) — only learned weights and the resulting spike rate/latency differ. That's exactly the portability claim this table should support in your paper.

3. Noise-Robustness Findings (Both Datasets)

UNSW-NB15

Depolarizing p	Accuracy	F1	FPR	ROC-AUC	F1 Retention
0.000	0.8575	0.8577	0.1713	0.9331	100.0%
0.005	0.8485	0.8429	0.1442	0.9321	98.3%
0.010	0.8455	0.8413	0.1559	0.9318	98.1%
0.020	0.8430	0.8328	0.1249	0.9286	97.1%
0.050	0.5165	0.0000	0.0000	0.9133	0.0%
0.100	0.5165	0.0000	0.0000	0.8594	0.0%

CIC-IDS2017

Depolarizing p	Accuracy	F1	FPR	ROC-AUC	F1 Retention
0.000	0.9690	0.9235	0.0255	0.9917	100.0%
0.005	0.9715	0.9295	0.0237	0.9923	100.7%
0.010	0.9680	0.9212	0.0268	0.9917	99.8%
0.020	0.9680	0.9179	0.0168	0.9910	99.4%
0.050	0.8345	0.2789	0.0000	0.9809	30.2%
0.100	0.8025	0.0000	0.0000	0.4713	0.0%

The failure mode is dataset-dependent — and that's a reportable result, not noise: both models hold up well through p=0.02 (97–100% F1 retention). Past that they fail differently. On UNSW-NB15 the model collapses hard at p=0.05, predicting one class for everything, while ROC-AUC stays at 0.91 — the underlying scores still rank correctly, only the decision threshold breaks. On CIC-IDS2017 the degradation is gradual at p=0.05 (F1=0.28, ROC-AUC still 0.98), then fails completely at p=0.1 where ROC-AUC itself drops to 0.47 (chance level) — here the underlying signal degrades, not just the threshold. Say explicitly in the paper that noise tolerance depends on the dataset, not just the architecture.

4. Code Changes to Make in Your Script

These are the fixes applied during the re-run. Apply them to your own copy before you run anything further — two of them silently dropped data in your original script.

Dataset loader — dropped the UNSW-NB15 test file HIGH — `qsnn/data/datasets.py`

Your file-finder was only picking up `training-set.csv` and silently skipping `testing-set.csv`, so every UNSW-NB15 result you've reported so far is from 175,341 rows instead of the full 200,000. Fix the glob/filename match so both official CSVs load, and verify with a row-count assertion (175,341 → 200,000 is the signal it's fixed).

Why it matters: this directly explains part of the accuracy/F1 shift between your draft and the re-run (Section 1) — it's not just sample-size noise.

CSV encoding — dropped one CIC-IDS2017 file HIGH — `qsnn/data/datasets.py`

One of the eight CIC-IDS2017 release CSVs has a stray non-UTF-8 byte that was failing to parse and getting silently skipped. Add a latin1 fallback when the default UTF-8 read raises `UnicodeDecodeError`, e.g.:
`pd.read_csv(path, encoding='utf-8', errors=...)` → wrap in try/except and retry with `encoding='latin1'`.

Why it matters: without this fix you're training on 7/8 of CIC-IDS2017 and won't know it.

Resumable/checkpointed training MEDIUM — `qsnn/engine.py, scripts/train.py`

Add a `--resume` flag and always save a `last.pt` checkpoint after every epoch (in addition to your existing best-F1 checkpoint). On start, if `--resume` is passed and `last.pt` exists, load it and continue from the saved epoch/optimizer state instead of starting over.

Why it matters: full training runs got interrupted repeatedly by background-process time limits. Without this, an interruption means losing all progress and restarting from epoch 0.

IBM hardware backend resolution MEDIUM — `qsnn/noise.py`

Passing the backend as a bare string (e.g. `"ibm_fez"`) fails silently or errors deep in the Qiskit call stack. Resolve it to a real backend object first: `service = QiskitRuntimeService(); backend = service.backend("ibm_fez")`, then pass that object through, not the string.

Why it matters: this is what let the real-hardware run get as far as "attempted, blocked by quota" instead of failing on a type error before it even reached IBM's queue.

Stratified noise-sweep subsample (new helper) LOW — add if you need speed — `scripts/make_noise_subsample.py`

New script: builds a stratified 2,000-sample subset of the test set (same class balance as the full set) for fast noise sweeps. The full 26,028-sample UNSW test set was taking 7+ hours per noise level — the subsample keeps the robustness trend representative while running in a practical time.

Use this if a full-size noise sweep is too slow for your available compute/time budget.

IBM hardware run script (new helper, ready but unused) LOW — run when quota resets — `scripts/run_ibm_hardware.py`

New script that runs a small sample through real IBM hardware via the fixed backend resolution above. It's ready to execute as soon as your IBM Quantum account's hardware-time quota is available again (it was exhausted at `ibm_fez` during this run, 0 queued jobs otherwise — a provider-side limit, not a bug).

This is how you close the "real hardware" item in your Limitations/Future Work once quota resets.

5. Things Worth Double-Checking Before You Cite Both Runs Together

- **Spike rate dropped by half** (28.4% → 14.4% on UNSW-NB15) despite near-identical detection metrics. Check: are `angle_gain`, `angle_bias`, and the firing threshold identical between your config and the re-run's `config.used.yaml`? If so, this is itself worth reporting (similar accuracy with much sparser firing on more data) rather than something to footnote away.
- **Random Forest baseline is noticeably lower in the re-run** (93.34% → 90.48% accuracy on UNSW-NB15). RF has no quantum-encoder stochasticity, so this almost certainly comes from the larger/different test distribution (official test partition now included), not from anything in your QSNN pipeline. State this plainly if both numbers appear in the same table.
- **Latency roughly doubled** (3.6–3.8ms → 7.9ms on UNSW-NB15). Expected — different machine, different PennyLane build. Report as-is, but note both are simulator latencies on different CPUs, not a like-for-like hardware comparison.

> **Methodological point to address, not patch over:** your draft's 50k-record, training-partition-only sample and the re-run's full 200k-record (official train+test combined) sample are not the same protocol. If both sets of numbers go in the same paper, say so explicitly — e.g. a dedicated "Reproducibility Check" paragraph or a table footnote — rather than presenting them as the same experiment at two scales. A reviewer who catches an unstated protocol change reads it as a bigger problem than one you disclosed yourself.

6. Paper Text to Update (Ready to Paste)

What your draft's own Limitations section (Section VI) already asked for — now addressed

RESOLVED *"Full noisy runs and real-hardware runs were not part of this experiment."* — the 6-point noise sweep on both datasets (Section 3) covers the noise half. Real hardware was attempted (`ibm_fez`) but blocked by exhausted account quota — state as "attempted, blocked by provider quota," not "not attempted."

RESOLVED *"Did not evaluate on... CIC-IDS2017."* — full pipeline run, 300,000 rows, all 8 release files, all 6 models (Section 2).

RESOLVED *"Quantum baselines (QSVC, QCNN) were only exercised on synthetic smoke-test data."* — both now have full results on both real datasets, fulfilling future-work item (v).

PARTIAL *"Results come from a single seed and a single split."* — still true. Multi-seed repeats were attempted but not completed in time. Keep this

limitation as-is, noted as in progress.

Add — New Subsection: Noise Robustness

Ready-to-paste text

We evaluated the trained QSNN-NIDS model under a depolarizing-plus-amplitude-damping noise channel at six levels (p = 0, 0.005, 0.01, 0.02, 0.05, 0.1), applied after every variational block, using PennyLane's density-matrix simulator. On UNSW-NB15, the model retains 97–98% of its ideal F1 through p=0.02, then collapses at p≥0.05 to predicting the majority class (F1=0), while ROC-AUC remains at 0.91–0.86 — indicating the underlying decision scores still separate the classes; only the fixed classification threshold fails under heavy noise. On CIC-IDS2017 the model is noise-tolerant through a wider range (99–101% F1 retention through p=0.02, and a graceful degradation to F1=0.28 at p=0.05 with ROC-AUC still 0.98), but fails completely at p=0.1, where ROC-AUC itself drops to 0.47 (chance level) — here the underlying signal degrades, not just the threshold. This dataset-dependent failure mode suggests that noise robustness in quantum IDS models should be characterized per-deployment rather than assumed to transfer.

Add — New Subsection: Cross-Dataset Evaluation (CIC-IDS2017)

Ready-to-paste text

To test whether QSNN-NIDS generalizes beyond UNSW-NB15, we repeated the full pipeline — identical architecture, hyperparameters and seed — on CIC-IDS2017 (Monday–Friday capture, 300,000 flows sampled across all eight release files, reduced via the same MI-filter-then-PCA pipeline to 8 features). The model reaches 97.46% accuracy, F1 0.937 and ROC-AUC 0.994, outperforming its own UNSW-NB15 numbers on every metric. Classical baselines retain their lead (Random Forest: 99.65% accuracy, F1 0.991), but the margin is narrower than on UNSW-NB15, and the model's ranking relative to every baseline is consistent across both datasets.

Revise — Limitations Section

Replace: "All circuits were simulated without noise… full noisy runs and real-hardware runs were not part of this experiment… We used 50,000 records… and did not evaluate on the official test partition or on other datasets such as CIC-IDS2017… Quantum baselines (a ZZ-kernel QSVC and a QCNN)… were only exercised on synthetic smoke-test data, so they are not reported here."

With

We additionally evaluated the model under simulated depolarizing/amplitude-damping noise (Section —) and on a second, independent dataset, CIC-IDS2017 (Section —), with full QSVC and QCNN baselines on both. A real-hardware run was attempted on IBM's ibm_fez backend but could not complete because the account's allotted quantum-processing time was exhausted at the time of the experiment; this remains open. Results throughout the paper are still from a single seed and a single split per dataset; multi-seed repeats to report confidence intervals are left for future work.

Revise — Future Work

- (i) run under realistic noise models and on IBM hardware → DONE (noise) · BLOCKED BY QUOTA (hardware)
- (iv) evaluate on CIC-IDS2017 with repeated seeds → DONE (single seed); repeated seeds still open
- (v) complete the comparison with QSVC and QCNN at full scale → DONE , both datasets

Remaining future work: (ii) learn the firing threshold and study other encodings; (iii) multi-class detection of individual attack families; (iv-cont.) repeated-seed confidence intervals and ToN-IoT; (vi) neuromorphic deployment for real energy measurement; and completing the real-hardware run once IBM quota is available.

7. Honest Limitations to Keep in the Paper

- **Single training seed (42) per dataset.** Mean±std across multiple seeds was attempted but not completed in the available time; reported numbers are single-run.
- **Real IBM quantum hardware:** attempted, blocked by quota. The IBM Quantum account's instance had exhausted its allotted hardware time (backend `ibm_fez`, 0 queued jobs otherwise) — a provider-side quota limit, not a code or scope gap. `scripts/run_ibm_hardware.py` is ready to run once quota is available.
- **No side-by-side table against your draft's original numbers yet** was in this document's precursor — now resolved, since Sections 1–2 above are exactly that table.